

- **Deadline:** Wednesday 2026-09-09 at 23:59
 - **To pass:** submit a serious attempt to answer the question.
 - **Hand in:** upload a PDF to the course webpage. Use 3-5 pages.
 - **Collaboration:** You work in groups of two.
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Your task is to investigate the ecology of two related butterflies and report on their differences. The butterflies are *Pieris rapae*, “small white”, and *Pieris napi*, “green-veined white”, two common butterflies in Sweden (and other parts of the world) where they are known as *rovfjäril* and *rapsfjäril*. As the latin name hints, these two butterflies are closely related and it is not surprising that they have similar ecology.

Both species are spread over most of the world: Europe, Asia, North America, and more. Both species hibernates, so you will see survivors appear in the spring. Larvae hatch in the summer.

The main question is: how are *P. rapae* and *P. napi* adapted to seasons?

- Use data from Artportalen, artportalen.se.
- How do you investigate this data?
- What scientific conclusions can you draw?
- How do you present your data?
- How do you present your conclusions?

Artportalen presents data collected by various research and community projects, as well as data collected by interested individuals (*citizen science*!) from all over Sweden for a range of species. You can visualize and download data in different ways.



Figure 1: *Pieris rapae* and *Pieris napi*.